Rank-Two Module SVP in Power-of-Two Cyclotomic Rings

Abstract. We prove that unit-threshold shortest vector decision is NP-complete for free rank-two modules over power-of-two cyclotomic rings in the coefficient maximum norm. Our deterministic reductions use basis coefficients of absolute value at most three. YES instances have first minimum one and NO instances at least two, giving search hardness below factor two. Sparse products at Sidon positions encode Boolean clauses. Reflection and positivity control extra collisions and unused inputs; a prime exponent excludes reciprocal factors. Geometric sums, parity, and an elementary reciprocal resultant give a principal coset. A rigid Littlewood polynomial homogenizes it. We include its construction using exact rational conditional expectations, a rational sampling grid, and deterministic coloring.

Keywords: Module lattices, Shortest vector problem, Cyclotomic rings, Deterministic reductions, Coefficient maximum norm

1 Introduction

Let $R_d = \mathbb{Z}[X]/(X^d+1)$, where d is a power of two. We study the shortest vector problem for rank-two modules over R_d in the coefficient maximum norm: the largest absolute coefficient of the two polynomial coordinates.

The main obstacle is closure under ring multiplication. A Boolean encoding must control every multiplier in R_d, including those that do not represent assignments.

1.1 Our result

The module rank can remain two, with every basis coefficient bounded by three. All degrees and coefficient arrays below are explicitly encoded; all hardness reductions are deterministic, polynomial-time, and many-one.

Theorem 1. *The following decision problems are NP-complete.*

1. *Given a nonsingular $G \in R_d^{2\times 2}$, decide whether GR_d^2 contains a nonzero vector of coefficient maximum norm at most one. Hardness holds for bases*

$$G = \begin{pmatrix} g & h \\ 0 & J \end{pmatrix}, \qquad g_i \in \{0,2\}, \quad h_i \in \{-1,0,1,3\}, \quad J_i \in \{-1,1\}.$$

The constructed lattice has first minimum one in the YES case and at least two in the NO case.

2. *Given $0 \neq g \in 2R_d$ and $h \in R_d$, decide whether $h + gR_d$ meets the coefficient unit cube. Hardness holds with the same restrictions on g, h as in the first part.*

Thus search approximation within any fixed factor $1 \leq \gamma < 2$ is NP-hard for these modules. On a YES instance, such an algorithm must return an integer vector of norm one; a NO instance contains no such vector. The lattice dimension grows as $2d$ while the module rank stays two. The reduction has large absolute constants, arising from the uniform construction of the polynomial J.

1.2 Overview of the proof

We pass through an affine bit equation

$$y = P_A x + c, \qquad x, y \in \{0,1\}^L, \qquad A = \mathbb{Z}[X]/(X^L + P),$$

where P is a sparse 0/1 polynomial and P_A denotes multiplication in A. Its encoding and its conversion to a module are separate constructions.

Boolean constraints in a sparse product. We express 3SAT by positive exactly-one constraints and place the input bits at explicit Sidon positions A_i. For each incidence of variable i in clause j, include $X^{U_j - A_i}$ in P, with the U_j widely separated. Unique nonzero differences isolate each clause sum from all other collisions. Two copies, with offsets zero and minus one, enforce exactly-one.

Reflecting the support about degree T makes P reciprocal. Unique unordered pair sums ensure that every new collision imposes only an already implied at-most-one constraint. Positivity in the output equations forces every unused input to zero; no input bits are prescribed. Choose a prime $p > 8T$ and set $L = (p+T)/2$. A common root of $w = X^L + P$ and its reciprocal would satisfy $\alpha^p = 1$. Positivity excludes one, while every other pth root has minimal degree $p - 1 > L$. Thus w has no reciprocal irreducible factor.

Packing and recovering equality. Complement the bits and pack the two sign words as

$$v = (S_L - 2y) + X^L(S_L - 2x), \qquad H = S_{2L} - 2c.$$

The affine equation is exactly $\mathrm{rem}_w(v - H) = 0$. In a larger dyadic ring, multiply $H + 2wu$ by a blockwise geometric sum. The identity $(1 - Z)S_k = 2$ and parity force a sign in each intended position and zero in every padding position, regardless of u.

An equality in R_D still leaves a congruence modulo $1 + X^D$ in A. We remove it by comparing integer algebraic norms. Newton's identities give a sufficient lower bound for the Mahler measure of each nonreciprocal factor of w; its reciprocal resultant controls all roots at once. A divided derivative handles repeated factors. This excludes every nonzero bounded remainder for $D \geq 16L^3$.

Homogenization. A sign polynomial J is rigid if its only nonzero multipliers giving bounded products are signed monomials. In a nonzero bounded vector $(gu + hv, Jv)$, rigidity forces $v = \pm X^j$; evenness of g excludes $v = 0$. Dividing by the monomial recovers a bounded coset point, which proves the reduction to rank-two SVP.

For completeness, we construct J deterministically. A flat Littlewood seed [1] bounds the Euclidean norm of every short-product multiplier, leaving polynomially many candidates. Conditional expectations on disjoint convolution rows select $O(\log d)$ coefficient flips excluding all of them. Every probability is an exact rational, and a linear flip penalty preserves the spectral lower bound. The seed and its rational coloring argument [2] are included in the appendices.

Organization. Section 3 proves the reduction and norm separation. Section 4 gives the sparse Boolean encoding, starting from 3SAT, and section 5 constructs the rigid polynomial. The standard background inputs are 3SAT NP-completeness, Bertrand's postulate, and Bernstein's inequality. A rational sampling grid makes the seed's interval selection explicit. All constructions specific to the reduction are proved in the paper.

2 Preliminaries

For dyadic d, write every element of R_d uniquely as $a = \sum_{j<d} a_j X^j$, using binary integer coefficients. Set $\|a\|_\infty = \max_j |a_j|$ and $\|a\|_2^2 = \sum_j a_j^2$. For $(a, b) \in R_d^2$, use $\|(a, b)\|_\infty = \max(\|a\|_\infty, \|b\|_\infty)$. All norms in this paper refer to these coefficient embeddings.

The polynomial $X^d + 1$ is irreducible over $\mathbb{Q}$: after substituting $X + 1$, Eisenstein's criterion applies at two. Indeed $(X + 1)^d \equiv X^d + 1 \pmod 2$ and the constant term of $(X + 1)^d + 1$ is two. Thus R_d is a domain. Multiplication by $\pm X^j$ is a signed permutation of coefficients and preserves both norms. Write $S_r = \sum_{j<r} X^j$ for every integer $r \geq 1$. In R_d we have

$$(1 - X)S_d = 2, \qquad S_d a \equiv a(1)S_d \pmod{2R_d}. \tag{1}$$

A nonsingular matrix $G \in R_d^{2\times 2}$ defines the free module $\Lambda = GR_d^2$. Its coefficient lattice has dimension $2d$ and integer basis obtained by multiplying each column of G by $1, X, \ldots, X^{d-1}$. Its first minimum is $\lambda_{1,\infty}(\Lambda) = \min_{0\neq z\in\Lambda} \|z\|_\infty$. Every such minimum is a positive integer.

For clarity, the decision languages reject malformed inputs. The principal-coset input lists d, g, h with dyadic d, $g \neq 0$ even and arbitrary $h \in R_d$. Its YES condition is the existence of $E \in \{-1, 0, 1\}^d$ with $E - h \in gR_d$. The module input lists a nonsingular 2×2 matrix over R_d. Its YES condition is $\lambda_{1,\infty} \leq 1$. All complexity bounds count bit operations in these explicit encodings.

3 From an affine bit equation to rank-two SVP

Our starting point is the affine equation

$$y = P_A x + c, \qquad x, y \in \{0,1\}^L, \qquad P_A x = \mathrm{rem}_w(Px), \quad w = X^L + P. \tag{2}$$

Here P has 0/1 coefficients, $P(0) = 1$, and degree $T < L/4$; $c \in \{-1, 0, 1\}^L$, and $L \geq 64$. Proposition 2 constructs such instances from 3SAT and ensures that w has no reciprocal irreducible factor. We first give the reduction using these properties; its Boolean source is proved in section 4.

Write $s = P(1)$. For every x with $\|x\|_\infty \leq 1$, the ordinary product $Px = d + X^L q$, with $\deg d < L$, has $\|d\|_\infty, \|q\|_\infty \leq s$. Because $\deg(Pq) < 2T < L$, one substitution gives

$$P_A x = d - Pq, \qquad \|P_A x\|_\infty \leq s + s^2. \tag{3}$$

This bound will control the remainder in the soundness proof.

3.1 Two sign words in a principal coset

The simple identity $(1 - Z)S_k = 2$ forces both signs and zero padding.

Lemma 1. *In $\mathbb{Z}[Z]/(Z^k + 1)$, if $\|S_k v\|_\infty \leq 1$, then $\|v\|_\infty \leq 1$. Consequently $v \equiv 1 \pmod 2$ forces $v = \pm 1$, whereas $v \equiv 0 \pmod 2$ forces $v = 0$.*

Proof. Set $E = S_k v$. Then $2v = (1 - Z)E$ and $\|v\|_\infty \leq \|E\|_\infty \leq 1$. The two conclusions follow coefficientwise from parity.

Let m be the least power of two at least $2L$, and choose a power of two $D \geq 16L^3$. Then $m \mid D$. Set

$$k = D/m, \qquad F = S_k(X^m), \qquad H = S_{2L} - 2c, \qquad g = 2Fw, \quad h = FH, \tag{4}$$

where c occupies the first L coordinates of H. The first L coefficients of H are in $\{-1, 1, 3\}$, its next L coefficients are one, and its remaining $m - 2L$ coefficients are zero. All products displayed in (4) have degree less than D, with disjoint coefficient blocks. In particular,

$$g \neq 0, \qquad g_i \in \{0, 2\}, \qquad h_i \in \{-1, 0, 1, 3\}.$$

Theorem 2 (Principal-coset reduction). *The construction (4) satisfies*

$$\exists x, y \in \{0,1\}^L : \; y = P_A x + c \quad \Longleftrightarrow \quad (h + gR_D) \cap \{-1, 0, 1\}^D \neq \varnothing.$$

It is deterministic and takes polynomial bit time. For the coset reduction, one can choose $D < 32L^3$.

Proof. Given a source solution, set $a = S_L - 2x$, $b = S_L - 2y$, and $v = b + X^L a$. These are two sign words. In $A = \mathbb{Z}[X]/(w)$,

$$\operatorname{rem}_w\left(\frac{v-H}{2}\right) = \operatorname{rem}_w(c - y - X^L x) = c - y + P_A x = 0.$$

Monic division therefore gives $v - H = 2wU$ for an integer polynomial U. Hence $h + gU = Fv$ is in the unit cube: its coefficients are repeated signs in the first $2L$ positions of each block and zeros elsewhere.

Conversely, suppose $E = h + gu$ has norm at most one, and set $v = H + 2wu \in R_D$. Write uniquely

$$v = \sum_{j<m} X^j v_j(X^m), \qquad v_j \in \mathbb{Z}[Z]/(Z^k + 1).$$

Multiplication by F acts separately on these blocks. Since $Fv = E$ and $v \equiv H \pmod 2$, lemma 1 gives $v_j = \pm 1$ for $j < 2L$ and $v_j = 0$ otherwise. Thus $v = b + X^L a$ for two sign words $a, b \in \{-1, 1\}^L$. Put $x = (S_L - a)/2$ and $y = (S_L - b)/2$. As R_D is torsion-free, we may cancel two in $v - H = 2wu$ and obtain

$$c - y - X^L x = wu \quad \text{in } R_D.$$

Lifting this identity to $\mathbb{Z}[X]$ and reducing modulo w gives

$$e := P_A x + c - y \in (1 + X^D)A.$$

The bound (3) gives $\|e\|_\infty \le s + s^2 + 2 \le 2s^2 < L^2$, since $s \le T + 1 < L/2$. By lemma 4, $e = 0$.

Choosing D as the least power of two at least $16L^3$ gives $D < 32L^3$. Every output coefficient is bounded by three, and all arrays are computed and written explicitly in polynomial time.

3.2 Separation by an integer resultant

The separation argument compares two algebraic norms. A bounded remainder has small norm, whereas $1 + X^D$ has large norm in every field component of the auxiliary quotient. A reciprocal resultant controls all roots at once; divided derivatives handle repeated factors.

For a monic polynomial p with roots $\alpha_1, \ldots, \alpha_d$, put

$$M(p) = \prod_i \max(1, |\alpha_i|).$$

When $p(0) = \pm 1$, write $p^{\#}(X) = X^d p(1/X)/p(0)$ for its normalized reciprocal.

Lemma 2. *Let $p \in \mathbb{Z}[X]$ be monic of degree d, with $p(0) = \pm 1$. If $p \ne p^{\#}$, then $\log M(p) \ge 1/(12d)$.*

Proof. Put $T = \log M(p)$. The power sums $s_m = \sum_i \alpha_i^m$ and $s_{-m} = \sum_i \alpha_i^{-m}$ are integers, since both p and $p^\#$ are monic integer polynomials. Newton's identities give $s_m \neq s_{-m}$ for some $1 \leq m \leq d$. Write $\alpha_i = e^{u_i + i\theta_i}$. Since $|p(0)| = 1$, $\sum_i |u_i| = 2T$ and $|u_i| \leq T$. Conjugation permutes the roots, so

$$1 \leq |s_m - s_{-m}| \leq \sum_i |e^{mu_i} - e^{-mu_i}| \leq 2me^{mT} \sum_i |u_i| = 4me^{mT}T.$$

If $T \geq 1/d$, the claim follows immediately. Otherwise $e^{mT} < 3$, and the displayed inequality gives $T \geq 1/(12d)$.

Lemma 3 (Reciprocal resultant). *Let $p \in \mathbb{Z}[X]$ be monic of degree d, with $p(0) = \pm 1$ and $\gcd(p, p^\#) = 1$. For every integer $D \geq 1$,*

$$\left|\mathrm{Res}(p, 1 + X^D)\right| \geq 2^{-d^2} M(p)^{D-2d}.$$

Proof. Write $r_i = |\alpha_i|$. The roots of $p^\#$ may be ordered as $1/\overline{\alpha}_i$. Their resultant is a nonzero integer, hence

$$1 \leq \left|\mathrm{Res}(p, p^\#)\right| = \prod_{i,j} \left|\alpha_i - \frac{1}{\overline{\alpha}_j}\right|.$$

For $i \neq j$, the corresponding factor is at most $2\max(1, r_i)\max(1, r_j^{-1})$. Since $\prod_i r_i = 1$, we also have $\prod_i \max(1, r_i^{-1}) = M(p)$. Bounding all off-diagonal factors gives

$$\prod_i |r_i - r_i^{-1}| \geq 2^{-d(d-1)} M(p)^{-2(d-1)}.$$

The reverse triangle inequality yields, for each root,

$$|1 + \alpha_i^D| \geq \max(1, r_i)^D \frac{|r_i - r_i^{-1}|}{2\max(r_i, r_i^{-1})}.$$

Indeed, for $r_i \geq 1$ use $1 - r_i^{-D} \geq 1 - r_i^{-1}$, and for $r_i \leq 1$ use $1 - r_i^D \geq 1 - r_i$. Multiplying and using $\prod_i \max(r_i, r_i^{-1}) = M(p)^2$ proves the claim.

Lemma 4 (Bounded-remainder separation). *Let $w \in \mathbb{Z}[X]$ be monic of degree $L \geq 64$, with $w(0) = \pm 1$, and suppose no irreducible factor of w is reciprocal. Put $A = \mathbb{Z}[X]/(w)$. If $D \geq 16L^3$ and $e \in \mathbb{Z}[X]$ satisfies*

$$\deg e < L, \qquad \|e\|_\infty \leq L^2, \qquad e \in (1 + X^D)A,$$

then $e = 0$.

Proof. Suppose $e \neq 0$, and lift the ideal-membership relation to

$$e = (1 + X^D)q + wu, \qquad q, u \in \mathbb{Z}[X].$$

Since $\deg e < L$, some irreducible factor p of w satisfies $j := \mathrm{ord}_p(e) < r := \mathrm{ord}_p(w)$. Let $d = \deg p$ and let α be a root of p. The constant term of p is ± 1.

Also $\gcd(p, p^{\#}) = 1$, since p is irreducible and nonreciprocal. In particular, p has no unit-circle root: such a root and its conjugate would give a common root of p and $p^{\#}$. Thus $p \nmid 1 + X^D$, and the lifted identity implies $\mathrm{ord}_p(q) = j$.

Put $e_j = e^{(j)}/j!$ and $q_j = q^{(j)}/j!$. Both are integer polynomials. Taking the jth divided derivative and evaluating at α, all lower derivatives of q and all derivatives of wu through order j vanish, leaving

$$e_j(\alpha) = (1 + \alpha^D) q_j(\alpha) \neq 0.$$

The nonzero algebraic integer $q_j(\alpha)$ has norm of absolute value at least one. Each coefficient of e_j is $\binom{i}{j} e_i$, of absolute value at most L^{j+2}. Since $(j+1)d \leq rd \leq L$, evaluating at the d roots of p gives

$$|N(e_j(\alpha))| \leq L^{(j+3)d} M(p)^{L-1} \leq L^{3L} M(p)^{L-1}.$$

Combining this with lemma 3 yields

$$2^{-d^2} M(p)^{D-2d} \leq |N(1 + \alpha^D)| \leq |N(e_j(\alpha))| \leq L^{3L} M(p)^{L-1}.$$

Since $D - 2d - L + 1 > 0$, taking logarithms and applying lemma 2 gives

$$\frac{D - 2d - L + 1}{12d} \leq d^2 \log 2 + 3L \log L.$$

The right side is less than $5L^2/4$, because $d \leq L$, $\log 2 < 1$, and $3 \log L \leq L/4$ for $L \geq 64$. The last inequality holds at 64 and its difference is increasing thereafter. The left side is at least $4L^2/3 - 1/4$ when $D \geq 16L^3$. This is a contradiction.

3.3 Homogenization in rank two

A rigid anchor is a sign polynomial $J \in R_D$ satisfying

$$\|Jv\|_\infty \leq 1 \quad \Longrightarrow \quad v = 0 \text{ or } v = \pm X^j \quad (0 \leq j < D).$$

Theorem 3 constructs one deterministically at every dyadic $D \geq 2^{3600}$.

Proposition 1 (Homogenization). *For $0 \neq g \in 2R_D$, any $h \in R_D$, and a rigid anchor J, put*

$$G = \begin{pmatrix} g & h \\ 0 & J \end{pmatrix}, \qquad \Lambda = G R_D^2.$$

Then Λ is a full-rank free rank-two module, and

$$(h + gR_D) \cap \{-1, 0, 1\}^D \neq \varnothing \quad \Longleftrightarrow \quad \lambda_{1,\infty}(\Lambda) = 1.$$

A NO coset gives $\lambda_{1,\infty}(\Lambda) \geq 2$.

Proof. Since R_D is a domain, $\det G = gJ \neq 0$. A bounded coset point $E = h + gu$ gives the norm-one vector (E, J). Conversely, let $(gu + hv, Jv)$ be nonzero of norm at most one. Rigidity gives $v = 0$ or $v = \pm X^j$. The first case makes gu even and bounded by one, hence zero, a contradiction. In the second case, multiplication by v^{-1} preserves the coefficient norm and gives $guv^{-1} + h$ in the source cube. The gap follows from integrality.

Proof (Proof of theorem 1). Combine the source in proposition 2 with theorem 2. For the module reduction choose

$$D = \max\left(2^{3600}, 2^{\lceil \log_2(16L^3)\rceil}\right).$$

The preceding construction applies at this degree. Construct the anchor of theorem 3 and apply proposition 1. The output is polynomial in the source size, since $D < 2^{3600} + 32L^3$, and it has the asserted coefficient restrictions.

Both languages are in NP. For the coset problem a certificate is $E \in \{-1, 0, 1\}^D$; expand multiplication by g into its negacyclic integer matrix, solve $gu = E - h$ over $\mathbb{Q}$, and check integrality. For the module problem use a nonzero $z \in \{-1, 0, 1\}^{2D}$ and solve $Gu = z$ in the same way. The integer matrices are nonsingular. Hadamard's determinant bound and exact rational elimination give polynomial bit complexity.

4 Boolean constraints in a sparse product

We encode an arbitrary Boolean formula directly in a sparse product. Distinct differences isolate the clauses. Reflecting the polynomial introduces only pairwise constraints already implied by those clauses; the resulting reciprocity also makes the algebraic separation elementary.

For a monic polynomial w of degree L, write $P_A a = \mathrm{rem}_w(Pa)$ in $A = \mathbb{Z}[X]/(w)$. As usual, a vector denotes its coefficient polynomial.

Proposition 2 (Sparse affine source). *There is a deterministic polynomial-time reduction from 3SAT to the following problem. The output consists of integers T, L, a reciprocal polynomial $P \in \{0, 1\}[X]$ of degree T, and $c \in \{-1, 0, 1\}^L$, with*

$$P(0) = 1, \qquad L \geq 64, \qquad L > 4T, \qquad w = X^L + P, \qquad \gcd(w, w^*) = 1.$$

The question is whether

$$y = P_A x + c \qquad \text{for some } x, y \in \{0, 1\}^L. \tag{5}$$

We give the construction and its proof. The source is positive exactly-one satisfiability: each clause requires exactly one of three distinct variables to be one. Its reduction from 3SAT is short. The clauses $(t, a, b), (t, a, d), (t, b, d)$ force $t = 1$ and $a = b = d = 0$: otherwise $a + b = a + d = b + d = 1$, which has no bit solution. Fix the resulting zero $f = a$. For each original variable introduce $x, \bar{x}$ and the clause $(x, \bar{x}, f)$. For an original disjunction $(\ell_1 \vee \ell_2 \vee \ell_3)$ introduce fresh $s_1, s_2, s_3, z_1, z_2, z_3$ and the clauses

$$(s_1, s_2, s_3), \qquad (s_i, \overline{\ell_i}, z_i) \quad (1 \leq i \leq 3).$$

The latter say $z_i = \ell_i - s_i$, or $s_i \leq \ell_i$. Thus one selector chooses a true literal, exactly when the disjunction is satisfied. All resulting clauses contain three distinct variables.

4.1 Isolating the incidences

Let the positive instance have v variables and r clauses. Choose an odd prime ℓ with $v < \ell < 2v$. Let $\rho_i \in \{0, \dots, \ell - 1\}$ be the residue of i^2 modulo ℓ, and set

$$A_i = 2\ell i + \rho_i \quad (1 \le i \le v), \qquad B = A_v. \tag{6}$$

The positions are increasing, with $B < 4v^2 + 2v$, and have distinct unordered pair sums. Indeed, $A_i + A_j = A_k + A_l$ forces $i + j = k + l$, since the difference of the residue sums has absolute value less than 2ℓ. Reducing modulo ℓ also gives $i^2 + j^2 = k^2 + l^2$. As ℓ is odd, the products ij and kl agree modulo ℓ. Thus the two pairs are the roots of the same quadratic over $\mathbb{F}_\ell$, and are equal as unordered pairs because all indices lie in $\{1, \dots, v\} \subset \{1, \dots, \ell - 1\}$. Distinct nonzero ordered differences follow by rearranging a difference equality as a pair-sum equality.

Duplicate each clause. Write C_j for the variable set in copy j, so $C_{2h-1} = C_{2h}$. Define

$$\Delta = 4B + 1, \qquad U_j = j\Delta \quad (1 \le j \le 2r), \qquad T = 2U_{2r} + 4B + 1, \tag{7}$$

and put

$$P = 1 + X^T + \sum_{j=1}^{2r} \sum_{i \in C_j} \left(X^{U_j - A_i} + X^{T - U_j + A_i} \right). \tag{8}$$

These exponents are distinct: different clause copies are separated by more than $4B$, and the two displayed families lie on opposite sides of $T/2$. Thus P has $0/1$ coefficients, degree T, weight $s = 2 + 12r$, and $X^T P(1/X) = P(X)$.

Choose a prime p with $8T < p < 16T$ and set

$$L = (p + T)/2, \qquad w = X^L + P. \tag{9}$$

Here T is odd, so L is an integer. Bertrand's postulate guarantees such a prime, which trial division finds in polynomial time in T. We have $L > 4T$ and $L < p - 1$.

Lemma 5 (The prime reciprocal exponent). *The polynomial w has no root on the unit circle and no reciprocal irreducible factor.*

Proof. Reciprocity of P gives $w^* = 1 + X^{L-T} P$. A common root α of w, w^* therefore satisfies

$$1 = \alpha^{2L-T} = \alpha^p.$$

It cannot be one, since $w(1) > 0$. Otherwise its minimal polynomial is Φ_p of degree $p - 1 > L$: irreducibility follows from Eisenstein's criterion at p applied to $\Phi_p(X + 1)$. This is again impossible. Thus w, w^* are coprime. Every unit-circle root of a real polynomial is also a root of its reciprocal, so none exists.

Let $\mathcal{A} = \{A_1, \dots, A_v\}$ and $\mathcal{F} = \mathcal{A} + \operatorname{supp}(P)$. Define the output translation by

$$c_z = \begin{cases} -1, & z = U_{2h} \text{ for some } 1 \le h \le r, \\ 0, & z \in \mathcal{F} \text{ otherwise}, \\ 1, & z \notin \mathcal{F}. \end{cases} \tag{10}$$

The negative case takes priority. Notice $\max \mathcal{F} = T + B < 2T$.

4.2 Forcing the unused inputs

Lemma 6 (Padding from positivity). *Every solution of* (5) *has* $x_z = 0$ *for* $z \notin \mathcal{A}$. *Consequently* $P_A x = Px$ *as ordinary polynomials.*

Proof. Write $Px = d + X^L q$ with $\deg d < L$ and $\deg q < T$. Then $P_A x = d - Pq$. Since $\deg(Pq) < 2T$, every coefficient at an index $i \geq 2T$ is the ordinary nonnegative coefficient of Px. Here $c_i = 1$, so the output bit condition forces that coefficient to be zero. Using the two terms $1, X^T$ of P gives

$$x_i = 0 \quad (2T \leq i < L), \qquad x_i = 0 \quad (T \leq i < L - T).$$

As $L \geq 3T$, these intervals cover $[T, L)$. In particular the last T input bits vanish, whence $q = 0$ and $P_A x = Px$ everywhere.

Apply positivity once more at every $i \geq T + B + 1$, also outside $\mathcal{F}$. The term X^T now forces $x_i = 0$ for $B < i < L - T$; together with the first step this gives $x_i = 0$ for every $i > B$. Finally every nonconstant exponent of P exceeds B. At $i \leq B$ the product coefficient is just x_i, and $\mathcal{F} \cap [0, B] = \mathcal{A}$. Thus $c_i = 1$ forces every remaining unused input to be zero.

4.3 Reading the clauses

Only the inputs $z_i = x_{A_i}$ remain. A contribution from the unreflected support belonging to copy j occurs at

$$U_j + A_k - A_i, \qquad i \in C_j, \quad 1 \leq k \leq v.$$

The row intervals for different copies are disjoint. Uniqueness of nonzero differences shows that each such row is a singleton except U_j, whose value is exactly $\sum_{i \in C_j} z_i$. The two copies of clause h therefore impose

$$\sum_{i \in C_{2h-1}} z_i \in \{0, 1\}, \qquad \sum_{i \in C_{2h}} z_i \in \{1, 2\},$$

so precisely one clause variable is one.

The reflected support belonging to copy j contributes at

$$T - U_j + A_i + A_k, \qquad i \in C_j, \quad 1 \leq k \leq v.$$

These intervals are disjoint from each other, the unreflected intervals, and the two guard families $A_i, T + A_i$. Uniqueness of unordered pair sums shows that a reflected row is either a singleton or a sum $z_i + z_k$ for distinct $i, k \in C_j$. Indeed a second representation must interchange i and k, so both are members of that clause. Its offset zero requires at most one of the two bits to be one, which is already implied by exactly-one.

Proof (Proof of Proposition 2). Soundness follows from the padding lemma and the two rows for each clause. Conversely, place a satisfying logical assignment

at the positions A_i and put every other input bit to zero. There is no overflow because $T + B < L$. At every ordinary or reflected singleton the product is a bit; every reflected pair sum is at most one. The two rows of each clause give outputs one and zero, respectively. All untouched rows have product zero and offset one. This defines a bit vector y satisfying the full equation.

The construction is polynomial: $B = O(v^2)$ and $T, L = O(rv^2)$.

5 Deterministic rigid anchors

Throughout this section, d is a power of two and $R_d = \mathbb{Z}[Z]/(Z^d+1)$. A Littlewood polynomial has all d coefficients in $\{-1, 1\}$. It is rigid if its only nonzero multipliers giving a product of coefficient norm at most one are the signed monomials.

Theorem 3 (Uniform rigid anchors). *For every power of two $d \geq 2^{3600}$, a deterministic algorithm outputs a rigid Littlewood polynomial in polynomial bit time in the explicit output length d.*

We begin with a Littlewood polynomial whose values at the roots of $Z^d + 1$ are at least a constant times $\sqrt{d}$. This restricts every short-product multiplier to constant squared length. We then change $O(\log d)$ coefficients to exclude all remaining multipliers except the signed monomials.

5.1 Screening the multipliers

Lemma 7 (Screening and parity). *Let f be Littlewood and suppose $|f(\xi)| \geq \gamma\sqrt{d}$ whenever $\xi^d = -1$. If $\|fv\|_\infty \leq 1$, then $\|v\|_2^2 \leq \gamma^{-2}$. If $v \neq 0$, then fv is Littlewood.*

Proof. For $\xi_\ell = \exp((2\ell + 1)\pi i/d)$, Fourier orthogonality gives

$$\|fv\|_2^2 = \frac{1}{d} \sum_{\ell<d} |f(\xi_\ell)|^2 |v(\xi_\ell)|^2 \geq \gamma^2 d \|v\|_2^2.$$

The left side is at most d. Modulo two, $f = S_d$ and $ZS_d = S_d$. Thus every coefficient of fv has the parity of $v(1)$, so a product of norm at most one is either zero or Littlewood. Since R_d is a domain and $f \neq 0$, the zero case implies $v = 0$.

Fix a rational constant $0 < c \leq 1$, and put $C = \lceil 4/c^2 \rceil$. We preserve a spectral bound of $(c/2)\sqrt{d}$. By lemma 7, it suffices to exclude

$$\mathcal{V} = \{v \in R_d : 2 \leq |\operatorname{supp} v|,\ \|v\|_2^2 \leq C\}.$$

Indeed, a one-term multiplier can give a bounded product only if it is a signed monomial. For $d \geq C$, the candidates are enumerable in $d^{O(C)}$ bit time, and $|\mathcal{V}| \leq d^{2C}$. To obtain the latter bound, encode v by $|v_j|$ copies of each signed index in a fixed order and pad to length C with a blank symbol. This uses $\|v\|_1 \leq \|v\|_2^2 \leq C$ and an alphabet of size $2d + 1 \leq d^2$.

Theorem 4 (Deterministic repair). *Let $d \geq C$ be a power of two, and put $\ell = \log_2 d$. Suppose a Littlewood polynomial f_0 satisfies $|f_0(\xi)| \geq c\sqrt{d}$ at every $\xi^d = -1$, and*

$$d \geq 16C^4\ell, \qquad 64C^3\ell \leq c\sqrt{d}.$$

A deterministic algorithm running in $d^{O(C)}$ bit time produces a rigid Littlewood polynomial.

Independently flip each coefficient of f_0 with probability p:

$$f = \sum_{j<d} f_{0,j}(1 - 2B_j)Z^j, \qquad \Pr(B_j = 1) = p.$$

For each candidate, we test a set of convolution rows that depend on disjoint flip flags. Their joint probability is an exact rational product, which makes conditional expectations elementary.

5.2 Independent tests and exact expectations

Lemma 8. *Fix $v \in \mathcal{V}$ with support size m, and suppose $0 < p \leq 1/(2C)$ is rational. One can deterministically select a set I_v of at least d/m^2 convolution rows that depend on pairwise disjoint sets of flip flags. Let E_v be the event that every selected row is a sign. Its probability, also conditional on any fixed prefix of the flags, is computable exactly as a rational number in polynomial time in d, 2^m, and the bit length of p. Moreover,*

$$\Pr(E_v) \leq \exp\left(-\frac{pd}{2C^2}\right).$$

Proof. Writing $S = \operatorname{supp} v$, row i depends on the flags $\{B_{i-j \bmod d} : j \in S\}$. It intersects at most $m(m-1)$ other row sets, one for each nonzero difference of two elements of S. Greedily retaining a row and deleting all intersecting rows therefore selects at least $d/[m(m-1)+1] \geq d/m^2$ disjoint rows.

Before the flips, each row is a sum of m nonzero integer terms $t_1, \ldots, t_m$. If their sum is not a sign, leaving all flags zero keeps it illegal, with probability at least $1 - mp \geq 1/2$. Otherwise their sum is $s \in \{-1, 1\}$. Flipping t_j alone preserves a sign exactly when $t_j = s$. At least one term differs from s, since otherwise their sum would be $ms \neq s$. Flipping only that term makes the row illegal, with probability $p(1-p)^{m-1} \geq p/2$. Thus every row is legal with probability at most $1 - p/2$. There are at least $d/m^2 \geq d/C^2$ selected rows. Their independence yields the stated exponential bound.

After fixing any prefix, the remaining flags are still independent. The selected row events remain independent, so their conditional joint probability is the product of the row probabilities. Each row probability is computed by summing over at most 2^m assignments. Products of at most d such rational probabilities have polynomial bit length.

Proof (Proof of theorem 4). Set

$$p = \frac{8C^3\ell}{d}, \qquad B = 16C^3\ell = 2pd.$$

The first dimension bound gives $p \le 1/(2C)$. Let T be the total number of flips, and use the events E_v from lemma 8. Consider the nonnegative random variable

$$W = \frac{T}{B+1} + \sum_{v \in \mathcal{V}} \mathbf{1}_{E_v}.$$

Its expectation is less than one:

$$\mathbb{E}W \le \frac{pd}{B+1} + d^{2C} e^{-pd/(2C^2)} < \frac{1}{2} + d^{-2C} < 1,$$

where $\ell \ge \log d$ gives the penultimate bound. After fixing a prefix σ of length r with $T(\sigma)$ ones, the conditional expectation is the exact rational number

$$\mathbb{E}[W \mid \sigma] = \frac{T(\sigma) + p(d-r)}{B+1} + \sum_{v \in \mathcal{V}} \Pr(E_v \mid \sigma).$$

At each step, compute its two branch values and choose the smaller. Their $(1-p, p)$ weighted average is the current value, so this procedure ends with $W < 1$. Consequently $T \le B$ and no event E_v occurs. In particular, no candidate has a Littlewood product.

Changing at most B coefficients changes any value on the unit circle by at most $2B$. The second dimension bound says $4B \le c\sqrt{d}$, hence

$$|f(\xi)| \ge |f_0(\xi)| - 2B \ge (c/2)\sqrt{d}.$$

By lemma 7, every nonzero bounded-product multiplier has squared length at most C and a Littlewood product. All candidates with at least two terms have been excluded; the remaining multipliers are signed monomials.

There are $d^{O(C)}$ candidates and d decisions. The selected rows, their probabilities, and both conditional expectations are computed exactly using rational numbers of polynomial bit length. Thus the total cost is $d^{O(C)}$ bit operations.

Proof (Proof of theorem 3). The spectral seed in theorem 5 has $c = 2^{-252}$. Take $C = 2^{506}$. At $d = 2^{3600}$, with $\ell = 3600 < 2^{12}$, we have

$$16C^4\ell < 2^{2040} < d, \qquad 64C^3\ell < 2^{1536} < 2^{1548} = c\sqrt{d}.$$

For $d = 2^k$, both d/k and $\sqrt{d}/k$ increase for $k \ge 3600$. The hypotheses of theorem 4 therefore hold at every larger dyadic degree. Apply that theorem to the seed.

References

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2. Levy, A., Ramadas, H., Rothvoss, T.: Deterministic discrepancy minimization via the multiplicative weight update method. In: Integer Programming and Combinatorial Optimization. Lecture Notes in Computer Science, vol. 10328, pp. 380–391. Springer (2017), https://arxiv.org/abs/1611.08752

A A deterministic flat seed

This appendix supplies the only analytic ingredient of the anchor construction. We include the construction to distinguish a uniform deterministic algorithm from a nonconstructive choice of one polynomial at each length. The real/imaginary-part method is due to Balister et al. [1]; the quantitative implementation below uses theorem 6.

Theorem 5. *For every dyadic $d \geq 2^{512}$, a deterministic algorithm, polynomial in d, constructs $f_0 \in \{-1,1\}^d$ such that*

$$|f_0(z)| \geq 2^{-252}\sqrt{d} \qquad (|z| = 1).$$

Here and below, a sign vector denotes its coefficient polynomial. We use the elementary Bernstein inequality $\|T^{(r)}\|_\infty \leq q^r \|T\|_\infty$ for a trigonometric polynomial of degree at most q. All suprema in this appendix are over real angles.

A.1 A real part with few small intervals

Put $N = d/4 - 1$ and define the Rudin–Shapiro pair

$$A_0 = B_0 = 1, \qquad A_{t+1}(z) = A_t(z) + z^{2^t} B_t(z), \quad B_{t+1}(z) = A_t(z) - z^{2^t} B_t(z).$$

Induction gives sign coefficients and $|A_t(z)|^2 + |B_t(z)|^2 = 2^{t+1}$ on the unit circle. Let t be the largest positive odd integer with $2^{t+11} + 2^t - 1 \leq 2^{-66}N$, and set

$$T = 2^{t+10}, \quad \gamma N = 2T + 2^t - 1, \quad \delta = 2^{-8}\gamma^{7/2}, \quad h = 30\,2^{(t+1)/2} 2^{-7} (2T/N)^3.$$

Then $2^{-69} < \gamma \leq 2^{-66}$ and $\delta > 2^{-250}$. With $z = \mathrm{e}^{2\mathrm{i}\theta}$ define

$$c(\theta) = \Re\big(z^T A_t(z) + z^{2T} B_t(z)\big),$$
$$s_e(\theta) = \Im\big(A_{<N+1}(z) - z^T A_t(z) - z^{2T} A_t(z)\big).$$

The notation $A_{<m}$ denotes the first m stable coefficients of the recursion. The two removed blocks of the stable sequence are both A_t: its coefficient at index $j = \sum b_r 2^r$ is $(-1)^{\sum b_r b_{r+1}}$, and the high digit at T or $2T$ is separated from the low t digits by zeros. Consequently the cosine frequencies of c and the sine

frequencies of s_e partition $\{2, 4, \dots, 2N\}$, each with sign coefficients. The norm identity and a decomposition of a prefix into aligned dyadic blocks give

$$|c| \le \sqrt{N}, \qquad |A_{<m}| < 5\sqrt{m}, \qquad |s_e| \le 6\sqrt{N}.$$

For the prefix bound, sum $\sum_{r=0}^{\lfloor \log_2 m \rfloor} 2^{(r+1)/2} < 5\sqrt{m}$.

Write $c(\theta) = C(\cos\theta)$, where C is an even integer polynomial of degree $q = 2\gamma N$. Put

$$R = 2\lceil 64N/h \rceil, \quad M = NR, \quad \eta = \min\{h/(128N^3), 1/(4M^2)\}.$$

Compute strictly decreasing rational $y_0, \dots, y_M$ with $y_0 = 1$, $y_M = -1$, $y_{M-r} = -y_r$, and $|y_r - \cos(r\pi/M)| \le \eta$; the bit complexity is given below. Mark a closed cell $[j\pi/N, (j+1)\pi/N]$ as bad if $|C(y_r)| \le 3h/4$ for some $jR \le r \le (j+1)R$. Extend with period π and merge consecutive marked cells into intervals $\mathcal{I}$ on the circle.

Lemma 9. *Every interval of $\mathcal{I}$ has length at most $6\pi/N$; distinct intervals are separated by at least π/N. The family is invariant under $\theta \mapsto \pi - \theta$ and $\theta \mapsto \pi + \theta$, and avoids the $100\pi/N$ neighborhoods of $\frac{\pi}{2}\mathbb{Z}$. There are $4J \le 4\gamma N$ intervals, of which J lie in $[0, \pi/2]$. Outside their union, $|c| > 5h/8 > \delta\sqrt{N}$.*

Proof. The signed cosine support gives $\|c'\|_\infty \le 2N^2$ and $|C'(y)| \le 8N^3$ for $|y| \le 1$. For the latter use the Chebyshev identity $T_j'(\cos\theta) = j\sin(j\theta)/\sin\theta$ and $|\sin(j\theta)| \le j|\sin\theta|$. Thus each marked cell contains a point with $|c| \le 13h/16 < h$. In an unmarked cell, comparison with the preceding fine-grid point gives

$$|c| > 3h/4 - 8N^3\eta - 2N^2\pi/M \ge 11h/16 - \pi h/64 > 5h/8.$$

Rescale $x = 2T\theta$, and put

$$A = 2^{-(t+1)/2} A_t(\mathrm{e}^{\mathrm{i}x/T}), \quad B = 2^{-(t+1)/2} B_t(\mathrm{e}^{\mathrm{i}x/T}), \quad F(x) = \Re(\mathrm{e}^{\mathrm{i}x} A + \mathrm{e}^{2\mathrm{i}x} B).$$

We have $|A|^2 + |B|^2 = 1$ and $\|A^{(r)}\|, \|B^{(r)}\| \le 2^{-10r}$. For $r \le 4$, the difference between $F^{(r)}$ and $E_r = \Re(\mathrm{i}^r \mathrm{e}^{\mathrm{i}x} A + (2\mathrm{i})^r \mathrm{e}^{2\mathrm{i}x} B)$ is less than $1/8$, by the product rule; also $|F^{(r)}| \le 2^r + 2$. Writing $a = \mathrm{e}^{\mathrm{i}x} A$, $b = \mathrm{e}^{2\mathrm{i}x} B$,

$$\Re a = (4E_0 + E_2)/3, \qquad \Re b = -(E_0 + E_2)/3,$$
$$\Im a = -(4E_1 + E_3)/3, \qquad \Im b = (E_1 + E_3)/6.$$

If all $|F^{(r)}| < 1/4$ for $0 \le r \le 3$, these identities imply $|a|^2 + |b|^2 < 55/64$, a contradiction.

The rescaled cell width is $a_0 = 2T\pi/N < 2^{-11}$. Seven consecutive marked cells would give points $x_0, \dots, x_6$, one in each cell, with $|F(x_j)| < a_0^3/128$. Here $h/2^{(t+1)/2} = (30/\pi^3)a_0^3/128 < a_0^3/128$. For $1 \le r \le 3$, interpolation at $x_0, x_2, \dots, x_{2r}$ and repeated Rolle's theorem give a point within $7a_0$ of x_0 where

$$|F^{(r)}| \le (r+1)r! a_0^{3-r}/128.$$

The derivative bound then gives $|F^{(r)}(x_0)| < 24/128 + 126/2048 = 255/1024 < 1/4$. Together with the value bound, this is a contradiction.

For odd t, the recursion at 1 and -1 gives $F(0) = F(T\pi) = 1$. Since $|F'| \le 4$ and $101a_0 < 1/8$, all cells within $100\pi/N$ of an axis are good. The even cosine frequencies give the stated symmetries. To count components, the endpoints ± 1 lie outside $K = \{y \in [-1,1] : |C(y)| \le 3h/4\}$, since $C(\pm 1) = 2^{(t+1)/2} > h$. Each component of K, including an isolated point, accounts for at least two zeros with multiplicity of $C^2 - (3h/4)^2$. Hence K has at most q components. Ordered sampling gives at most q marked runs; each run marks a consecutive range of coarse cells. The period-π extension therefore has at most $2q = 4\gamma N$ components. The identity $y_{M-r} = -y_r$ and the evenness of C give reflection symmetry. Axis avoidance makes every symmetry orbit have size four. Finally direct substitution gives

$$\frac{h}{\delta\sqrt{N}} = \frac{30}{16}\left(\frac{2T}{\gamma N}\right)^{7/2} > \frac{30}{16}(2048/2049)^{7/2} > 8/5.$$

A.2 Filling the small intervals

Assign signs $\alpha(I)$ to the J intervals in $[0, \pi/2]$, extended by $\alpha(\pi - I) = \alpha(I)$ and $\alpha(\pi + I) = -\alpha(I)$. For $K > 0$, define

$$b_{2j-1} = K\sqrt{N}\int_{-\pi}^{\pi} \sum_{I\in\mathcal{I}} \alpha(I) 1_I(u) \sin((2j-1)u)\, du,$$

$$s_b(\theta) = \sum_{j=1}^{N} b_{2j-1} \sin((2j-1)\theta).$$

Lemma 10. *For any choices of the interval signs and $N \ge 2^{20}$, $|s_b| \le 5K\sqrt{N}$ everywhere and $|s_b| \ge (2K/3)\sqrt{N}$ on $\bigcup \mathcal{I}$.*

Proof. Reflection and summation of the odd-frequency progression give, for $\theta \in [0, \pi/2]$,

$$\frac{s_b(\theta)}{K\sqrt{N}} = \sum_{I\subset[0,\pi/2]} \alpha(I) \int_I \left(\frac{\sin(2N(u-\theta))}{\sin(u-\theta)} - \frac{\sin(2N(u+\theta))}{\sin(u+\theta)}\right) du.$$

For monotone g on a grid interval $[a, b]$, integration by parts using a sine primitive vanishing at both endpoints gives

$$\left|\int_a^b g(u)\sin(2Nu+\phi)\, du\right| \le |g(b) - g(a)|/N.$$

Disjoint intervals are bounded by total variation. Apply this first to $1/\sin(u+\theta)$ on either side of its minimum, then to $1/\sin v - 1/v$ on $[-\pi/2, \pi/2]$. The latter is increasing and has total variation less than two: differentiate, using $\sin^2 v \ge$

$v^2 \cos v$. Finally, intervals at distance at least π/N from θ contribute at most $2/\pi$ to the kernel $\sin(2Nv)/v$. All discarded contributions therefore have absolute value at most

$$E_N = \frac{2}{N \sin(100\pi/N)} + \frac{12\pi + 2}{N} + \frac{2}{\pi} < \frac{2}{3}.$$

The term $12\pi/N$ allows an interval crossing the minimum of $1/\sin(u+\theta)$; its length is at most $6\pi/N$ and the function is at most two there.

For completeness, let $\mathrm{Si}(x) = \int_0^x \sin v/v \, dv$. On a grid interval containing θ, the remaining local integral is $\mathrm{Si}(2N(b-\theta)) + \mathrm{Si}(2N(\theta-a)) \in [4/3, 4]$. On an interval not containing θ, it lies in $[-1, 2]$. Indeed, successive unsigned half-period integrals strictly decrease, so $0 \le \mathrm{Si}(x) < 2$ for $x \ge 0$ and $\mathrm{Si}(x) \ge \mathrm{Si}(2\pi) > 4/3$ for $x \ge \pi$. For an inside interval, the two nonnegative arguments sum to at least 2π; one is at least π. For an outside interval, the integral is $\mathrm{Si}(B) - \mathrm{Si}(A)$ with $B \ge A + 2\pi$. The bounds follow from $4/3 < \mathrm{Si}(2\pi)$ and $\mathrm{Si}(\pi) < 2$. These two inequalities follow by the alternating series $\sum_{r \ge 0} (-1)^r x^{2r+1}/((2r+1)(2r+1)!)$. When θ lies in an interval, it is the only one closer than π/N; otherwise at most two intervals are that close. Thus the lower bound is $4/3 - E_N > 2/3$ and the upper bound is $4 + E_N < 5$. An empty family causes no exception.

A.3 Two deterministic rounding steps

Take $K = 9000$. If $J > 0$, apply theorem 6 to the N rows with J columns

$$V_{j,I} = 4K\sqrt{N} \int_I \sin((2j-1)u) \, du, \qquad c_* = 4\sqrt{\ln(64N/J)}.$$

The entropy budget is $N\mathrm{e}^{-c_*^2/16} = J/64$, and $\|V_j\|_\infty \le 24\pi K/\sqrt{N}$. The resulting interval signs satisfy

$$|b_{2j-1}| \le (240c_* + 2400)\sqrt{J/N}\, 24\pi K < 3/4.$$

For the last inequality use $J/N \le 2^{-66}$; the expression is increasing on that range and $c_* < 29$ at its right endpoint. If $J = 0$, set all $b_{2j-1} = 0$.

Round these fractional coefficients to signs by another application of theorem 6. There are N columns, and rows indexed by $1 \le k \le 16N$, $0 \le \ell \le N$:

$$Q_{k,\ell,j} = \left(\frac{2j-1}{2N}\right)^\ell \sin((2j-1)\theta_k + \ell\pi/2),$$
$$\theta_k = \frac{(2k-1)\pi}{64N}, \qquad c_\ell = 4\sqrt{(11+\ell)\ln 2}.$$

Entries have magnitude at most one and the entropy sum is at most

$$16N \sum_{\ell \ge 0} 2^{-(11+\ell)} = N/64.$$

Writing the output as $s_o(\theta) = \sum_j \epsilon_{2j-1} \sin((2j-1)\theta)$, we obtain, allowing the small rational errors described below,

$$\frac{|(s_o - s_b)^{(\ell)}(\theta_k)|}{(2N)^\ell} \le (5281 + 240\ell)\sqrt{N}.$$

For $\ell > N$ the trivial bound $2N$ suffices instead. Every point of $[0, \pi/2]$ is within $\pi/(64N)$ of a sample. Taylor expansion, followed by reflection, now gives

$$|s_o - s_b| \le \sqrt{N} \sum_{\ell \ge 0} (5281 + 240\ell) \frac{(\pi/32)^\ell}{\ell!} = (5281 + 15\pi/2)\mathrm{e}^{\pi/32}\sqrt{N} < 5900\sqrt{N}.$$

By lemma 10, $|s_o| > 100\sqrt{N}$ on the bad intervals.

Define the Laurent polynomial

$$\begin{aligned} L(\mathrm{e}^{\mathrm{i}\theta}) &= 1 + 2c(\theta) + 2\mathrm{i}(s_e(\theta) + s_o(\theta)), \\ f_0(w) &= w^{2N} L(w) + w^{4N+1} + w^{4N+2} + w^{4N+3}. \end{aligned}$$

Every coefficient of L at exponents $-2N, \ldots, 2N$ is a sign: cosines give equal pairs and sines give opposite pairs. Thus f_0 has exactly $d = 4N + 4$ sign coefficients. Off the bad intervals its real part in the Laurent representation is at least $2\delta\sqrt{N} - 1$ in magnitude; on them the imaginary part is at least $2(100 - 6)\sqrt{N}$. Multiplication by w^{2N} preserves magnitude and the three appended terms cost at most three. Since $N > d/5$ and $d \ge 2^{512}$, $\delta\sqrt{N} > 1$, so $2\delta\sqrt{N} - 4 \ge \delta\sqrt{N} - 3$; also $\delta\sqrt{N} - 3 \ge 2^{-250}\sqrt{d/5} - 3 \ge 2^{-252}\sqrt{d}$. This proves the claimed lower bound.

Exact computation and bit complexity. None of the rounding steps needs an oracle for real numbers. Replace every matrix entry by a rational within $\varepsilon = N^{-20}$ and every threshold by a rational upper enclosure within ε. In the first rounding, the resulting error is bounded by

$$240\varepsilon\sqrt{J}(24\pi K/\sqrt{N} + \varepsilon) + (240c_* + 2400)\sqrt{J}\varepsilon + J\varepsilon < 1/1000.$$

Recompute the resulting b to coordinate error ε, giving a rational starting point in $[-4/5, 4/5]^N$ for the second rounding. In that step the errors are absorbed by the constant 5281 above. Rational enclosures for sine, cosine, square roots and logarithms follow from Taylor series with factorial tails, bisection, and

$$\begin{aligned} \pi &= 16 \arctan(1/5) - 4 \arctan(1/239), \\ \ln u &= 2 \sum_{r \ge 0} \frac{((u-1)/(u+1))^{2r+1}}{2r+1} \qquad (1 \le u \le 2). \end{aligned}$$

These procedures take polynomial bit time at the required precision.

For the rational grid, $h \ge N^{-3}$ gives $M \le 128N^5 + 2N$ and $\eta^{-1} = N^{O(1)}$. Compute the first half of the cosine approximations by Taylor series, set $y_0 = 1$ and $y_{M/2} = 0$, and reflect for the other half. Adjacent exact cosine values differ by

at least $2\sin^2(\pi/(2M)) \geq 2/M^2 > 2\eta$, so the rationals are strictly ordered. The integer Chebyshev recursion computes C with polynomial degree and coefficient bit length. All marking comparisons are exact rational comparisons, including equality cases. This takes polynomial bit time. The second coloring has $16N(N+1)$ rows and N columns; together with theorem 6, this completes the deterministic polynomial-time construction in theorem 5.

B Rational deterministic coloring

We give the finite-bit rounding tool used in appendix A. The argument is a rational implementation of the weighted partial-coloring method of Levy, Ramadas and Rothvoss [2]. All rationals are encoded by binary numerators and denominators.

Lemma 11. *Let $r \geq 64$, $V \in \mathbb{Q}^{M\times r}$, $x_0 \in [-1,1]^r \cap \mathbb{Q}^r$, and $\lambda_i \in \mathbb{Q}_{\geq 0}$ satisfy $\sum_i \mathrm{e}^{-\lambda_i^2/16} \leq r/64$. In deterministic polynomial bit time one can find $y \in [-1,1]^r \cap \mathbb{Q}^r$ with at least $\lceil r/2 \rceil$ sign coordinates and*

$$|\langle V_i, y - x_0\rangle| \leq 68\lambda_i \|V_i\|_2.$$

Proof. Delete zero rows and rows with $\lambda_i^2 > 4r$, whose required inequality holds for every point of the cube. Choose a dyadic q_i with $\|V_i\|_2 \leq q_i < 2\|V_i\|_2$, and retain both signs of $a_i = V_i/q_i$. Write m for the resulting number of rows. Their norms are at most one, and $\sum_i \mathrm{e}^{-\lambda_i^2/16} \leq r/32$. If $m = 0$, round directly to signs. Otherwise put

$$\Delta = (16r)^{-1}, \quad T_* = 8r/\Delta^2, \quad k_* = \lceil r/8 \rceil, \quad \eta = 2^{-\lceil \log_2(1024m(T_*+1))\rceil}.$$

Start at $x = x_0$ with dyadic weights $\mathrm{e}^{-\lambda_i^2} \leq w_i \leq \mathrm{e}^{-\lambda_i^2} + \eta$.

Until half the coordinates are signs, let I index the k_* largest weights (or all rows if fewer), and set

$$b_i = 4\Delta^2\lambda_i^2/r, \quad \rho_i = 1 - b_i, \quad h = \sum_i w_i\rho_i\lambda_i a_i, \quad M_0 = \sum_i w_i\lambda_i^2 a_i a_i^T.$$

Let U be the rational subspace zero at fixed coordinates and orthogonal to h, to a_i for $i \in I$, and to every a_i with $\lambda_i \leq 1$. The last group has fewer than $r/16$ members, by the entropy budget. Thus $\dim U \geq r/4$. Compute the orthogonal projector P_U by rational elimination. Among its nonzero columns $q_j = P_U e_j$, select one minimizing $q_j^T M_0 q_j/\|q_j\|_2^2$. Since $\sum_j q_j q_j^T = P_U$, this minimum is at most

$$\frac{\mathrm{tr}(P_U M_0)}{\mathrm{tr}\, P_U} \leq \frac{4}{r}\sum_i w_i\lambda_i^2.$$

Divide the selected column by a dyadic number between its norm and twice its norm, obtaining z with $1/2 < \|z\|_2 \leq 1$. Orient z so that $\langle x, z\rangle \geq 0$, and take

the largest rational $0 < \alpha \le 1$ with $x' = x + \Delta\alpha z$ in the cube. Put $y = \alpha z$, $t_i = \Delta\lambda_i\langle a_i, y\rangle$ and update

$$x \leftarrow x', \qquad w_i \leftarrow \eta \left\lceil \frac{w_i\rho_i(1 + t_i + t_i^2)}{\eta} \right\rceil .$$

A full step increases $\|x\|_2^2$ by more than $\Delta^2/4$; a truncated step fixes a new coordinate. Hence there are fewer than $4r/\Delta^2 + r < T_*$ steps. Orthogonality to h cancels the linear change in the total weight; the choice of direction bounds the quadratic change by $(4\Delta^2/r)\sum_i w_i\lambda_i^2$, canceled by ρ_i. Thus upward rounding alone increases the total weight, and at all times

$$\Phi = \sum_i w_i \le r/32 + (T_* + 1)m\eta < r/16.$$

A row outside I has weight less than $\Phi/k_* < 1/2$; its next weight is at most $3/2 + \eta$. A row inside I has $t_i = 0$ and increases by at most η. Looking at its last step outside I (or its initial value) shows that every final weight is less than two.

For $\lambda_i \le 1$ the discrepancy is zero. Otherwise $1 + t + t^2 \ge \mathrm{e}^t$ for $|t| \le 1$ and $1 - b \ge \mathrm{e}^{-2b}$ for $0 \le b \le 1/2$ give

$$w_i^{\mathrm{final}} \ge \exp\bigl(\lambda_i\langle a_i, x - x_0\rangle - (33 + 8\Delta^2)\lambda_i^2\bigr).$$

Together with $w_i^{\mathrm{final}} < 2$, both row signs, and $q_i < 2\|V_i\|_2$, this proves the bound 68.

All comparisons are rational. The weights lie on a fixed dyadic grid below two. The rational constraints defining U have polynomial bit length independent of the denominator of x; determinant bounds give a uniform polynomial bound for each direction's bit length. Writing $x_j = m_j/D$ and $z_j = n_j/E$, a full step adds only polynomially many denominator bits. If coordinate j causes truncation at $\epsilon \in \{-1, 1\}$, then

$$x_i' = x_i + \frac{(\epsilon D - m_j)n_i}{Dn_j},$$

so a truncated step has the same property. There are $O(r^3)$ steps. Finally e^{-s}, $s = \lambda_i^2 \le 4r$, is enclosed to accuracy $\eta/2$ by a rational Taylor approximation to e^s followed by inversion. Taking $L \ge \max(32r, \lceil \log_2(16/\eta)\rceil)$ terms gives a tail at most $2s^L/L! \le 2^{1-L}$. This proves the asserted bit complexity.

Theorem 6. *For $r \ge 1$, $V \in \mathbb{Q}^{M\times r}$, $x_0 \in [-1, 1]^r \cap \mathbb{Q}^r$, and rational $c_i \ge 0$ with $\sum_i \mathrm{e}^{-c_i^2/16} \le r/64$, one can deterministically find $x \in \{-1, 1\}^r$ such that*

$$|\langle V_i, x - x_0\rangle| \le (240c_i + 2400)\sqrt{r}\|V_i\|_\infty.$$

The bit complexity is polynomial in the input length.

Proof. If $r \le 64$, round to nearest signs; the error is at most $r\|V_i\|_\infty \le 8\sqrt{r}\|V_i\|_\infty$. Otherwise apply lemma 11, keep $\lceil r/2\rceil$ fixed coordinates, and recurse on the

other $s = \lfloor r/2 \rfloor$ coordinates, with starting point the current fractional values and thresholds $c_i + 4$. The new budget is valid because $\sum_i \mathrm{e}^{-(c_i+4)^2/16} \leq \mathrm{e}^{-1} r/64 \leq s/64$. The discrepancy bound follows by induction from

$$68c_i + \frac{240(c_i + 4) + 2400}{\sqrt{2}} \leq 240c_i + 2400.$$

There are logarithmically many levels; the denominator bounds in the previous proof are additive across them.